

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

APPLICATION TO RECOMMENDER SYSTEMS

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Collaborative Filtering

Details and images from Koren, Bell, Volinsky (2009).

Recommender systems: matching customers with products.

- Given: data on prior purchases/interests of users
- Recommend: further products of interest

Prototypical example: Netflix.

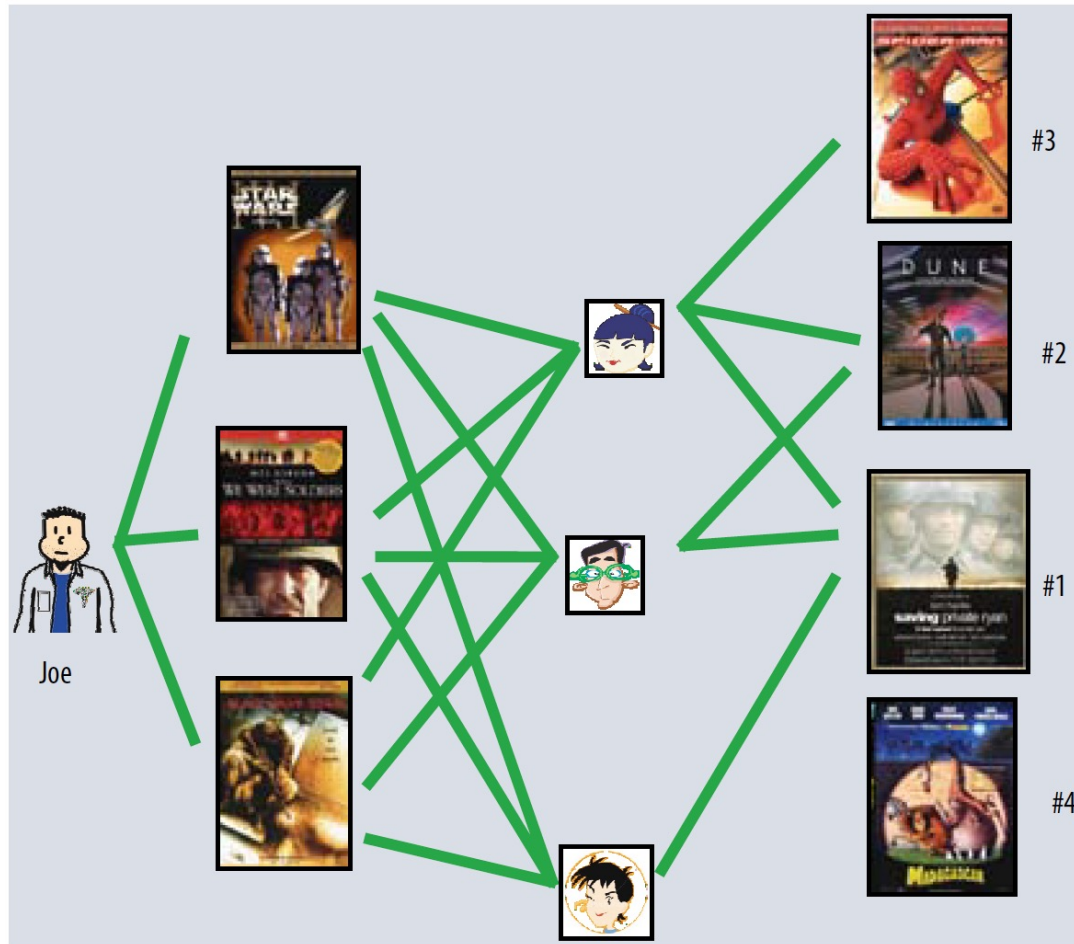
A successful approach: **collaborative filtering**.

- Model dependencies between different products, and between different users.
- Can give reasonable recommendations to a relatively new user.

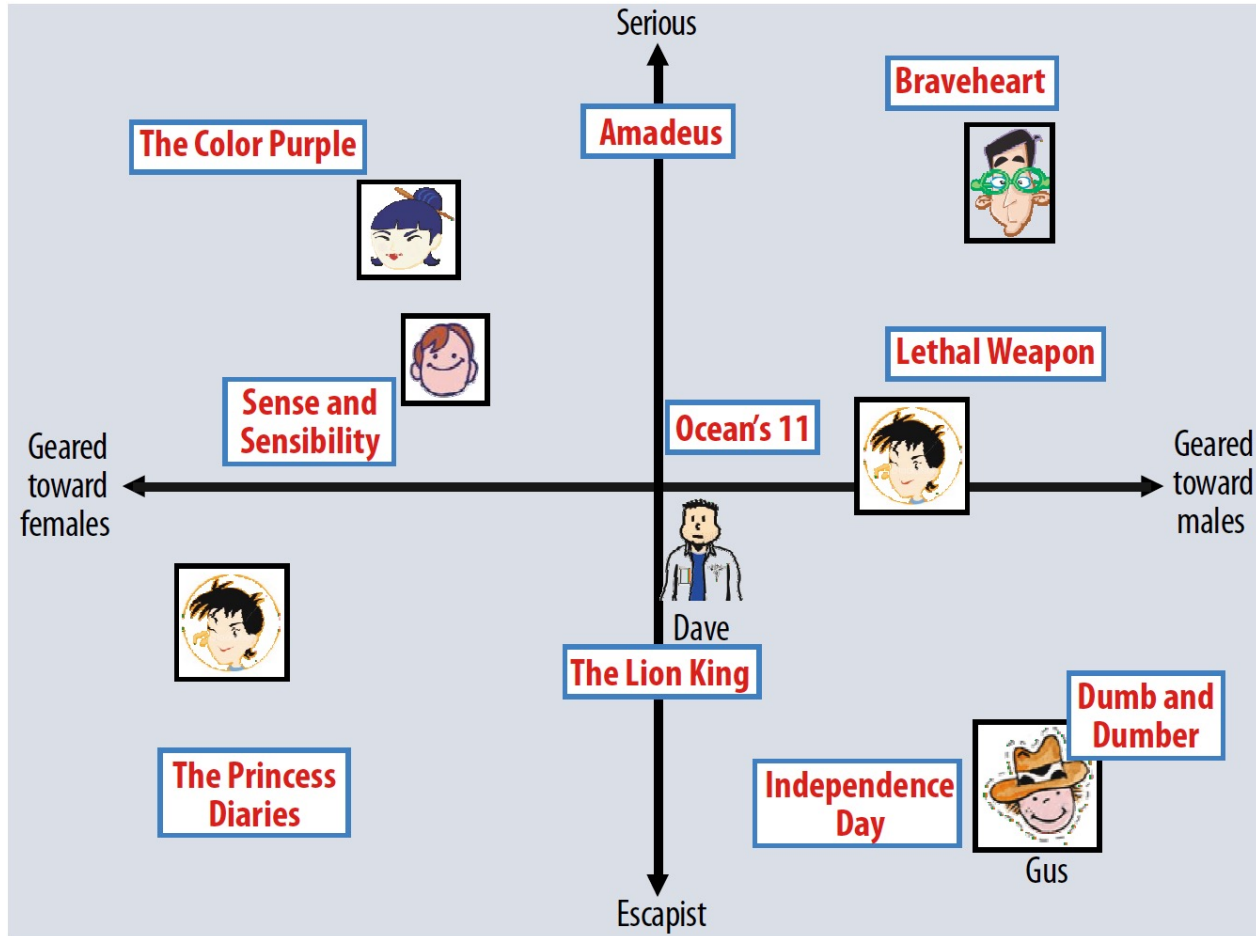
Two strategies for collaborative filtering:

- Neighborhood methods
- Latent factor methods

Neighborhood Methods



Latent Factor Methods



The Matrix Factorization Approach

User ratings are assembled in a large matrix M :

	Star Wars	Matrix	Casablanca	Camelot	Godfather	...
User 1	5	5	2	0	0	
User 2	0	0	3	4	5	
User 3	0	0	5	0	0	
	⋮					

- Not rated = 0, otherwise scores 1-5.
- For n users and p movies, this has size $n \times p$.
- Most of the entries are unavailable, and we'd like to predict these.

Idea: Get best low-rank approximation of M and use to fill in the missing entries.

Latent Semantic Indexing, Cont'd

Best rank- k approximation is of the form $M \approx UW^T$:

$$\underbrace{\begin{pmatrix} \longleftrightarrow \text{user 1} \longrightarrow \\ \longleftrightarrow \text{user 2} \longrightarrow \\ \longleftrightarrow \text{user 3} \longrightarrow \\ \vdots \\ \longleftrightarrow \text{user } n \longrightarrow \end{pmatrix}}_{n \times P \text{ matrix } M} \approx \underbrace{\begin{pmatrix} \longleftrightarrow u_1 \longrightarrow \\ \longleftrightarrow u_2 \longrightarrow \\ \longleftrightarrow u_3 \longrightarrow \\ \vdots \\ \longleftrightarrow u_n \longrightarrow \end{pmatrix}}_{n \times k \text{ matrix } U} \underbrace{\begin{pmatrix} \uparrow w_1 & \uparrow w_2 & \cdots & \uparrow w_p \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}}_{k \times p \text{ matrix } W^T}$$

Thus user i 's rating of movie j is approximated as

$$M_{ij} \approx u_i \cdot w_j$$

User and Movie Factors

$$\underbrace{\begin{pmatrix} \longleftrightarrow \text{user 1} \longrightarrow \\ \longleftrightarrow \text{user 2} \longrightarrow \\ \longleftrightarrow \text{user 3} \longrightarrow \\ \vdots \\ \longleftrightarrow \text{user } n \longrightarrow \end{pmatrix}}_{n \times P \text{ matrix } M} \approx \underbrace{\begin{pmatrix} \longleftrightarrow u_1 \longrightarrow \\ \longleftrightarrow u_2 \longrightarrow \\ \longleftrightarrow u_3 \longrightarrow \\ \vdots \\ \longleftrightarrow u_n \longrightarrow \end{pmatrix}}_{n \times k \text{ matrix } U} \underbrace{\begin{pmatrix} \updownarrow w_1 \updownarrow & \updownarrow w_2 \updownarrow & \cdots & \updownarrow w_p \updownarrow \end{pmatrix}}_{k \times p \text{ matrix } W^T}$$

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"Latent" representation embeds users and movies in the same k -dimensional space:

- Represent i th user by $u_i \in \mathbb{R}^k$
- Represent j th movie by $w_j \in \mathbb{R}^k$

Top Two Netflix Factors

